

Complex Analysis – 2

1 сентября 2026 года

Теоретический минимум.

Опр. A function is *entire* if it is holomorphic on all of $\mathbb{C}$. The whole sheet is about *rigidity*: a holomorphic function is determined by very little data, and any extra normalisation usually forces it to be a constant or a Möbius map.

Теорема 1 (Лиувилль и его усиления) *A bounded entire function is constant; if $|f(z)| \leq C(1+|z|)^m$, then f is a polynomial of degree at most m . Малая теорема Пикара: a non-constant entire function omits at most one complex value.*

Теорема 2 (лемма Шварца и Шварц–Пик) *If $f: \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic with $f(0) = 0$, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality only for $f(z) = e^{i\theta}z$. In general $|\frac{f(z)-f(w)}{1-\overline{f(w)}f(z)}| \leq |\frac{z-w}{1-\overline{w}z}|$, and the automorphisms of $\mathbb{D}$ are exactly $e^{i\theta} \frac{z-a}{1-\overline{a}z}$.*

Теорема 3 (принцип максимума и лемма Адамара о трёх прямых) *$|f|$ attains its maximum only on the boundary; if f is bounded and holomorphic on the strip $0 < \operatorname{Re} z < 1$ and continuous up to the boundary, then $|f(z)| \leq A_0^{1-\operatorname{Re} z} A_1^{\operatorname{Re} z}$, where A_j is the supremum of $|f|$ on the line $\operatorname{Re} z = j$.*

Факт 1 Nulls of a bounded holomorphic function on $\mathbb{D}$ satisfy the Blaschke condition $\sum(1 - |a_n|) < \infty$; finite Blaschke products $\prod \frac{z-a_j}{1-\overline{a_j}z}$ are precisely the holomorphic maps of $\overline{\mathbb{D}}$ onto itself taking the circle to the circle.

1. (Высшая лига 2017) Does there exist a non-constant entire function f with $(f(z))^3 = 1 + e^z$ for all $z \in \mathbb{C}$?
2. (Высшая лига 2018) Let p, q, r be non-constant nowhere vanishing entire functions with $p + q + r = 0$. Does there exist an entire function h such that p, q and r are constant multiples of h ?

3. (Высшая лига 2026) Can the zero set of a bounded holomorphic function on the upper half-plane $H = \{z : \operatorname{Im} z > 0\}$ be equal to $\{ki : k \in \mathbb{N}\}$?
4. (ISTCiM 2025, team tour) Let $\Omega = \{z \in \mathbb{C} : -1 < \operatorname{Re} z < 1\}$ and let S be the set of analytic functions $f : \Omega \rightarrow \mathbb{C}$ with $|f(z)| < 1$ on Ω and $f(0) = 0$. Evaluate $\sup_{f \in S} |f(i)|$.
5. (IMC 2021) Let $D \subseteq \mathbb{C}$ be an open set containing the closed unit disc, let $f : D \rightarrow \mathbb{C}$ be holomorphic and let p be a monic polynomial. Prove that $|f(0)| \leq \max_{|z| \leq 1} |f(z)p(z)|$.